

HR ATTRITION ANALYTICS & EMPLOYEE RISK PREDICTION REPORT

Executive Summary

This project analyzed employee attrition patterns and developed a machine learning model to identify employees at risk of resignation. Using SQL, Python, and Business Intelligence tools, workforce trends were examined across compensation, satisfaction, tenure, and overtime dimensions. The analysis provides actionable insights to support retention strategies and workforce planning.

Introduction

Employee attrition impacts productivity, recruitment costs, and organizational stability. The objective of this analysis was to identify the primary factors influencing employee turnover and build a predictive model capable of estimating employee resignation risk.

Background

The analysis was conducted using the IBM HR Analytics Employee Attrition dataset containing 1,470 employee records and 35 workforce attributes including demographics, compensation, satisfaction scores, job roles, and employment history.

Purpose

- The purpose of this project was to:
- Analyze historical attrition patterns.
- Identify key workforce retention drivers.
- Predict employee resignation probability.
- Support proactive HR intervention and workforce planning.

Method

The project followed a structured analytics workflow:

1. SQL-based data exploration and feature engineering.
2. Python exploratory data analysis and preprocessing.
3. Machine learning model development and evaluation.
4. Employee risk scoring and segmentation.
5. Dashboard development for executive reporting.

Sample

The sample consisted of 1,470 employees from various departments including Sales, Research & Development, and Human Resources.

Results

Key Workforce Metrics

- Total Employees Analyzed: 1,470
- Employees Who Left: 237
- Overall Attrition Rate: 16.1%
- High-Risk Employees Identified: ~170
- Departments Analyzed: 3
- Job Roles Analyzed: 9

Key Findings

- Employees working overtime exhibited significantly higher attrition rates.
- Lower-income employees demonstrated elevated resignation risk.
- Job Satisfaction and Work-Life Balance emerged among the strongest retention factors.
- Early-tenure employees showed higher attrition probability than long-tenured employees.
- Sales-related roles recorded the highest attrition concentration.

Predictive Modeling

- Classification Models Evaluated: Logistic Regression, Random Forest, XGBoost
- Best Performing Model: Random Forest
- Generated individual employee risk scores ranging from 0%–100%
- Employees segmented into Low, Medium, and High Risk categories for HR intervention.

Recommendations

- Prioritize retention programs for high-risk employees.
- Monitor overtime workloads and employee well-being.
- Improve engagement initiatives targeting low-satisfaction employees.
- Review compensation strategies for lower-income workforce segments.
- Implement proactive retention plans for newly hired employees.

Summary

This project demonstrates how data analytics and machine learning can transform workforce data into actionable business insights. By combining SQL, Python, predictive modeling, and executive reporting, the solution enables organizations to proactively reduce attrition risk and improve employee retention outcomes.

References

IBM HR Analytics Employee Attrition Dataset | Python | SQL | Scikit-Learn | Looker Studio

Conclusion

The analysis identified several critical factors influencing employee attrition, including overtime, monthly income, job satisfaction, work-life balance, and employee tenure. Through the examination of 1,470 employee records, the project uncovered workforce segments with elevated resignation risk and provided data-driven insights into employee retention challenges.

The machine learning model successfully generated individual employee risk scores, enabling proactive identification of employees most likely to leave. By combining descriptive analytics, predictive modeling, and executive reporting, this solution demonstrates how organizations can transition from reactive attrition management to proactive workforce planning.

Overall, the project highlights the value of leveraging analytics and machine learning to improve employee retention, reduce recruitment costs, enhance workforce stability, and support strategic HR decision-making.